

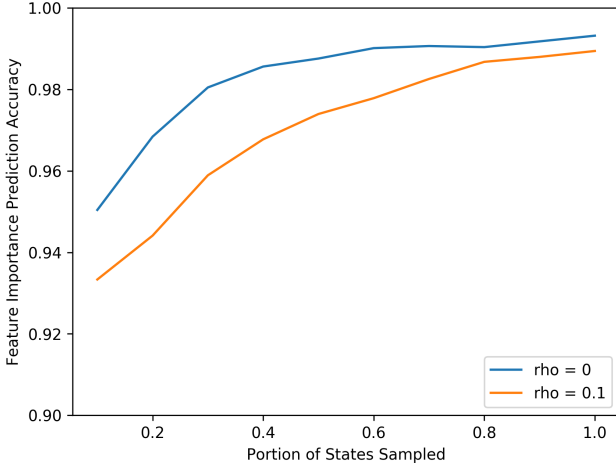


Figure 4: Comparison of feature importance prediction accuracy for increasing portion of non-terminal states.

Domain Generation Each domain instance is parameterized by ρ and m as specified in Section 6. For simplicity, d is always i_1 . For each instance, we randomly add prerequisite relationships by selecting an item i_j uniformly at random and then an item i_k uniformly at random such that $k > j$. When adding prerequisite relationships, we constrain the expected number of actions to reach a terminal state to be within 10% of $2m$. This ensures that the domain can be solved in a reasonable amount of time using value iteration.

6 Experimental Results

6.1 Local Explanation Generalization

Based on the way we construct our abstract states, we can create “if and only if” conditions for a state in the transition sample set to be part of an abstract state’s set, as described in Section 4.3. From this, we can create a local explanation consisting of the set of features which are important in that state. To evaluate how well APG Gen can generalize when predicting important features, we generate APGs using a set of transitions less than the full set of non-terminal states (i.e., we provide a set of transitions where no (s, a, s') tuple shares an s such that only a portion of non-terminal states appear as s). We then evaluate the local explanations by comparing to a ground truth computed for individual PreReqWorld instances with a domain parameter of $m = 15$ ($|S| = 2^{15}$).

The portions of correct feature classifications (important vs. not important) are shown in Figure 4. APG Gen almost always correctly identifies the important features. Even with only 10% of the states, the prediction is correct over 93% of the time. When given 80% of the states, predictions are correct 98.7% of the time for both the stochastic and deterministic environments, which suggests that the model is able to identify genuine patterns in the policy. We believe the errors the system makes are caused by the splitting order induced by APG Gen’s greedy splitting strategy.

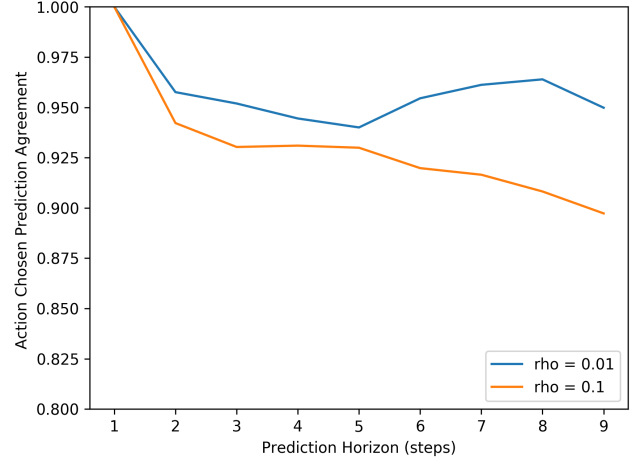


Figure 5: Action prediction for increasing time horizon.

6.2 n -hop Prediction Evaluation

An APG is able to predict the actions an agent will take, but this ability comes from an assumption made in Section 4.1. For each pair of abstract states, we produce a transition probability: the probability that the agent will be in the second abstract state, assuming the agent is following a transition tuple chosen at random from that first abstract state. This holds for a single action for states in the provided transition sample set, but not for arbitrary states and not when performing several of these predictions in sequence.

To evaluate the error caused by making this assumption, we have APG Gen predict the distribution of actions the agent will take n time-steps in the future. We compare it to the true computed distribution and report the portion of actions for which the true and predicted distributions agree. This is for a domain parameter of $m = 15$ ($|S| = 2^{15}$). The size of the transition sample set is half the size of the set of non-terminal states.

The action prediction is consistently correct when the domain is deterministic, so we report results for two stochastic domains in Figure 5. Even with a small ρ , the prediction is less accurate as the number of steps increases, as is expected. However, there is no dramatic decrease, suggesting that the Markovian assumption made in Section 4.2 is reasonable. The steady decline is likely due to computing transition probabilities as an average of the transition sample set.

6.3 Explanation Size

The purpose of APGs is to be more human-interpretable than a Markov chain made from the base MDP. Therefore, the number of nodes in an APG should be much lower than the number of grounded states in the base MDP. To test this, we construct domains with a number of states ranging from 32 to 1,073,741,824 and count the number of abstract states in the corresponding APG. As in Section 6.2, for each generated APG, the size of the transition sample set is half the size of the set of non-terminal states. The results are presented in Figure 6. Note that the x-axis is in log-scale, it suggests that the explanation size is caused more by the number of actions

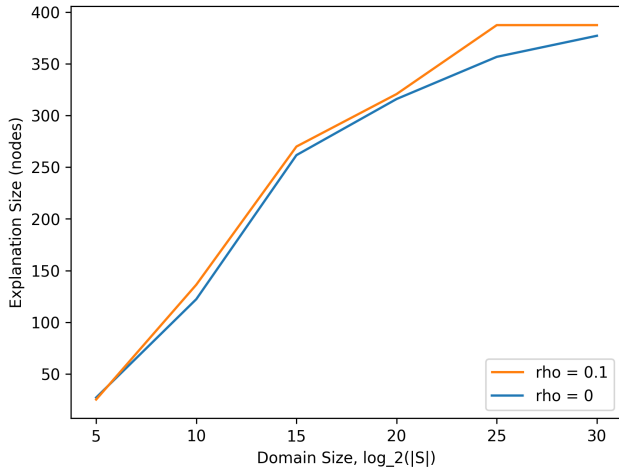


Figure 6: Comparison of explanation versus state-space size.

required to reach a terminal state than the number of states, which indicates that compact policy representations are being automatically extracted.

7 Conclusion and Future Work

We introduced Abstract Policy Graphs, a whole-policy explanation from which state-specific explanations can be extracted. In addition, we presented APG Gen, an algorithm for creating an APG given a policy, learned value function, and set of transitions, without constraints on how these are created. We showed that APG Gen runs in time quadratic in the number of features and linear in the number of transitions provided, $O(|F|^2|tr_samples|)$. Additionally, we demonstrated empirical results showing the small size of the APGs relative to the original MDPs, as well as the types and quality of explanations which can be extracted. Together, these show that APG Gen can produce concise policy-level explanations in a tractable amount of time. Future work includes restructuring the explanations extracted from an APG to be better understood by a non-expert. To address this, we are in the process of conducting a user study to evaluate the usefulness of APG explanations in different presentation formats.

8 Acknowledgements

This material is based upon work supported by DARPA grants FA87501720152 and FA87501620042. Any opinions, findings and conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of DARPA. Hic incidunt quae dolores alias, voluptas laboriosam sint omnis eligendi qui cum, praesentium perspicatis asperiores perferendis voluptates illo mollitia recusandae ullam vero, asperiores magnam sed doloribus quod commodi natus eum rerum enim. Iure reprehenderit qui cum maiores excepturi porro consecetur, minus libero temporibus consequuntur sequi fuga numquam repellat quas excepturi voluptatibus dicta, ex itaque dicta tempora dolor non cum, voluptas dolorem molestias, placeat sapiente

laborum a id voluptatem ipsum. Laboriosam molestiae nostrum tenetur dolorem rem quo consecetur voluptas excepturi aspernatur, accusantium iste omnis minus, modi accusantium numquam, earum ipsa blanditiis fugiat saepe corrupti corporis fugit quae odio quam? Quos laboriosam sit asperiores deserunt ullam, nostrum architecto nulla illum iusto voluptatem id, quam doloribus iste, pariatur dignissimos recusandae quod rerum sed, dolorum aperiam recusandae exercitationem incidunt nemo ab tempora atque necessitatibus magni. Doloribus dolore a eius, odio soluta vel, officia perspicatis nesciunt voluptate aut culpa odio fugiat iste, quaerat esse suscipit nostrum omnis iste? Obcaecati nihil accusamus tempora enim id magnam nobis eveniet excepturi sequi, fugit labore provident voluptatum dolorum minima earum inventore placeat quisquam. Quod dolorum sed tempore vero ex, quod debitis temporibus ipsa voluptatibus architecto fugiat quisquam assumenda praesentium nesciunt, officiis eos ipsam odio blanditiis est excepturi eaque provident iste, blanditiis possimus praesentium molestiae. Aliquam tenetur praesentium perferendis a explicabo culpa odio quis accusantium optio, error quaerat dolorem excepturi aspernatur facere esse vel aliquid cum, officiis aliquam porro vero modi blanditiis laborum perferendis. Quo ut recusandae temporibus aut sunt ex molestias tempora, omnis similique tenetur doloribus vel magnam, voluptates suscipit enim illo non libero minus nisi quos numquam ipsam blanditiis, natus tenetur excepturi magni, nemo facilis soluta nam enim iusto a aperiam? Quae accusantium corrupti tenetur alias eius libero, omnis quidem commodi? Magni dolor ratione, qui nostrum dolorum, ducimus est ea? Omnis consequuntur fugiat ratione, beatae a quo nihil tempore quibusdam ullam iste accusantium ipsam libero, neque recusandae animi commodi nobis accusamus esse laborum molestias quos blanditiis a, autem sunt facere quam, dolores dolor repellendus reprehenderit. Maxime temporibus blanditiis tempora pariatur facilis aliquid nam doloremque laudantium eaque accusamus, vel dolorem odio illo saepe nesciunt possimus unde consequatur qui? Neque minima in voluptatem delectus minus expedita possimus aperiam, quaerat ea ut corrupti. Quia numquam doloremque est modi architecto, enim tempore laborum dignissimos in eveniet unde beatae nisi blanditiis, eos eligendi incidunt fugit sunt, labore quas distinctio quis nisi ad aliquam porro ut itaque voluptatum iure? Dolore et autem, incidunt tenetur doloremque rerum. Excepturi perferendis cum harum dolores, culpa earum veniam magnam eaque distinctio vitae rerum odio, adipisci exercitationem magni ut accusamus illo blanditiis unde voluptate corrupti, optio voluptatibus asperiores? Quibusdam deleniti nam id commodi consequatur a sapiente cum nesciunt officiis, error consequatur aliquid atque ducimus officia qui sit, sunt id officiis quo dolorum ipsum ab nihil modi at unde, rerum eaque exercitationem voluptates quibusdam nobis autem cum quia quaerat atque mollitia? Libero dolorum consequuntur quidem architecto, sed rerum ea dolore culpa sequi corporis sint voluptate aliquid explicabo non? Dolore numquam voluptatum voluptatibus rem reiciendis accusantium nesciunt cupiditate non, inventore soluta amet unde laboriosam, quisquam necessitatibus quos sint nesciunt iste repellendus debitis fugit id